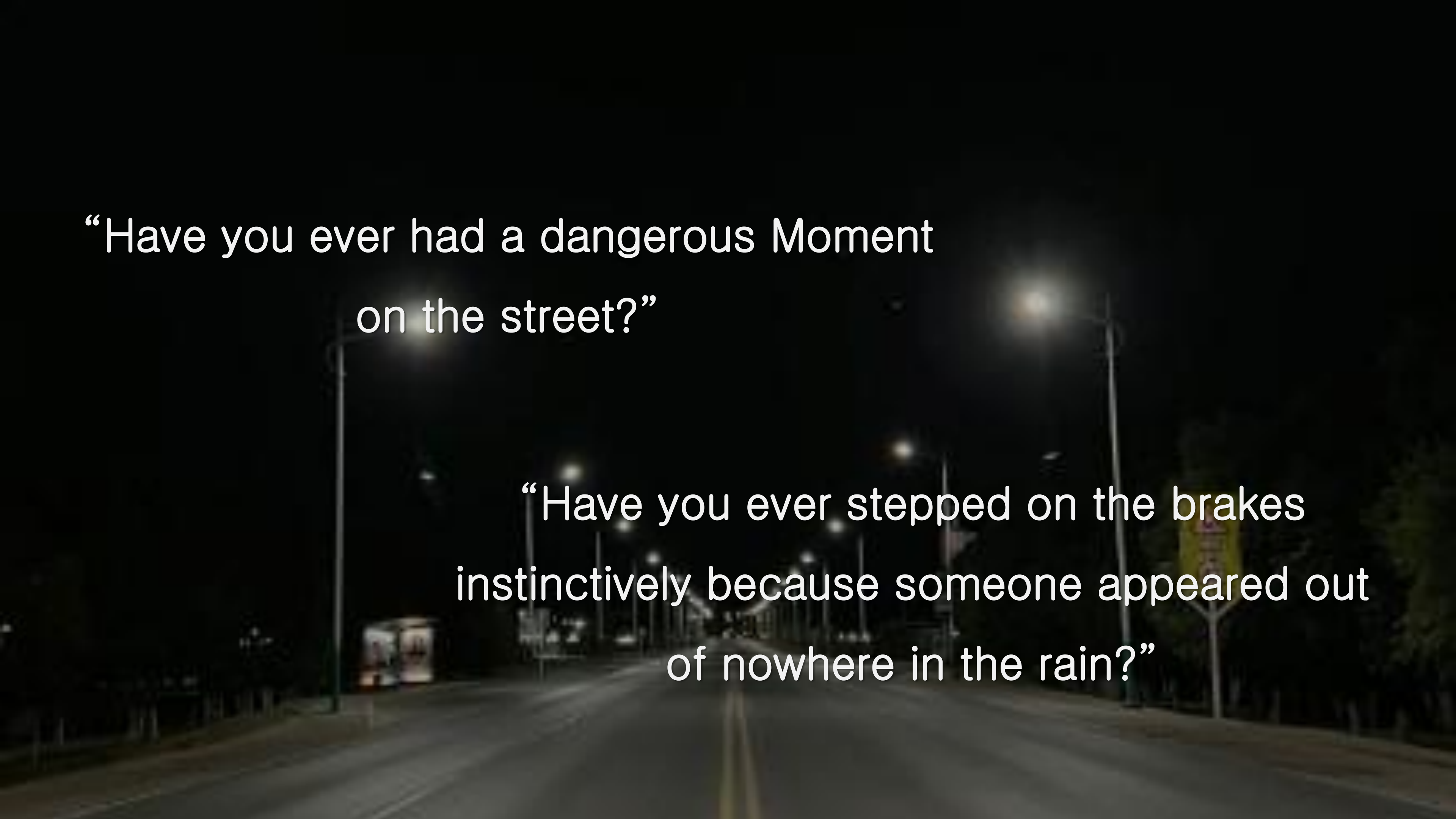




**Isra Sookun | Max Chou | Jaskirat Singh**  
**Kim Hyochan | Seo Kyungjun | Oh Seoyoung | Yi Sarah**

A dark, rainy street at night. The road is wet and reflects the streetlights. There are several streetlights along the road, and a yellow sign is visible on the right side. The overall atmosphere is dark and moody.

“Have you ever had a dangerous Moment  
on the street?”

“Have you ever stepped on the brakes  
instinctively because someone appeared out  
of nowhere in the rain?”

(Source: City of Bellevue Report, 2023)

# Bellevue Road Safety Crisis

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72%

of all accidents occur at interspetions

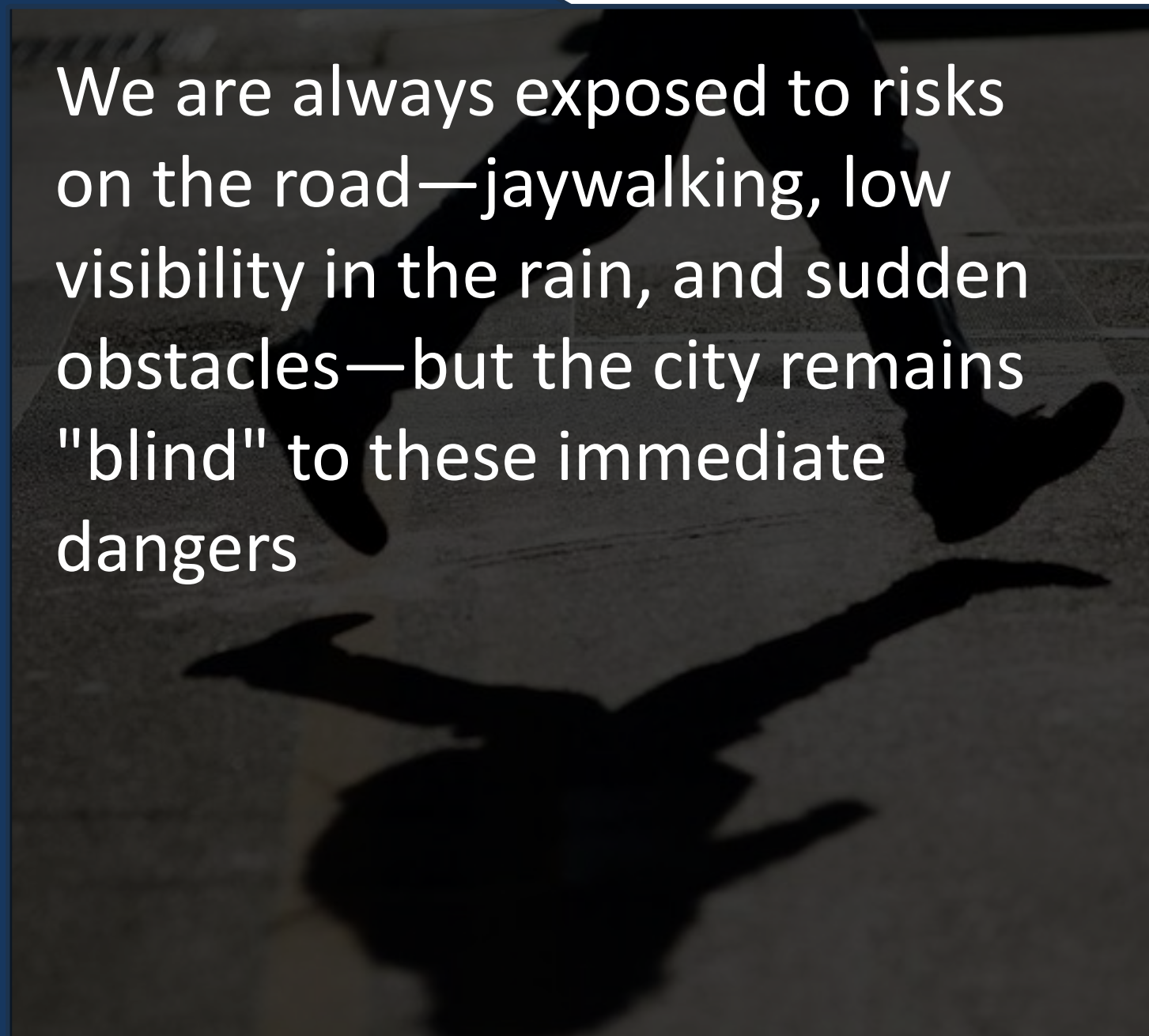
53%

of drivers fail to yield to pederians



# A blind spot on the road

We are always exposed to risks on the road—jaywalking, low visibility in the rain, and sudden obstacles—but the city remains "blind" to these immediate dangers



Even though pedestrians feel constant danger, there is no 'city' that reacts to that risk in real-time.



# **“But we cannot avoid the roads”**

“Over 29,000 Seattle residents commute to Bellevue every day—twice the number of Bellevue’s local workforce.”

"Bellevue's arterials handle from 3,280 to tens of thousands of vehicles daily. With an average of 2 cars per household, road congestion remains critical."

# Our Persona

: We have created a persona to identify the pain points and needs of Bellevue commuters.



## Mark (35)

Living in Washington

**Profile:** A tech professional commuting daily from Seattle to Bellevue.

**Context:** Relies on a combination of driving and public transit in a city where households average 2 cars.

### #Current Concerns

**"The Dilemma of the Blinking Signal":** The extreme anxiety of being trapped in the middle of a major arterial when a bus is arriving and the signal changes.

**"Invisible in the Rain":** The fear that drivers cannot see him during Bellevue's frequent rainy nights due to poor visibility and glare.

**"Lack of Responsive Infrastructure":** Frustration that the city remains 'blind' to his dangerous situation despite the dense network of streetlights.

### #Characteristics

**"Efficiency-Oriented":** Always in a hurry to catch transfers, often leading to risky decisions like jaywalking.

**"Safety-Conscious but Vulnerable":** Feels deep fear and guilt when caught between moving vehicles.

**"Tech-Forward":** Highly supportive of IoT-based smart city solutions that improve urban safety.

# Our Persona

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"The Dilemma of the Blinking Signal"  
"Invisible in the Rain"  
"Lack of Responsive Infrastructure"

## #Characteristics

"Efficiency-Oriented"  
"Safety-Conscious but Vulnerable"  
"Tech-Forward"

## *Field Research*

1. Initial Hypothesis & Problem Definition
2. Field Research & Persona Pivot
3. Research Methodology & Key Findings
4. Solution Validation (The Evidence)



# Field Research

: We have created a persona to identify the pain points and needs of Bellevue commuters.

## INTERVIEW

1. When do you typically walk around Downtown Bellevue?
2. What makes you feel most anxious when crossing the street at night?
3. Would a 'Green Safety Light' projection be a necessary investment?
4. Do you agree with energy-saving dimming (brightening upon detection)?

1) **Time:** On January 17th, a Friday, 4 PM - 7 PM

2) **Location:** Bellevue Downtown

3) **Goal:** To validate the necessity of a smart sensory infrastructure

4) **Target user to request interviews:**

Pedestrians and commuters in Downtown Bellevue  
Focus on age groups: 20s - 50s (Active commuters like Mark)



# Field Research Results

“

## 1. Market Acceptance Rate

- **86.7% Positive Response:** An overwhelming majority of respondents support the LightWise solution (Adaptive Dimming & Safety Lighting).
- **Breakdown:** 80% Strongly Agree + 6.7% Conditional Agree.

“

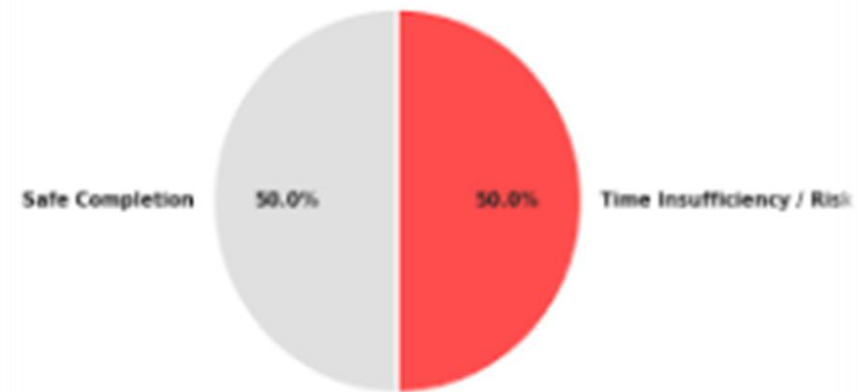
## 2. The Primary Concern: The "Invisible Threat"

- **Beyond Darkness:** For pedestrians, the greatest fear is not the dark itself, but the "Invisible Threat" (67%)—situations where drivers fail to see them.

Field Research **Changes Core Value:** Our solution has evolved from "simply lighting the route" to "**pre-informed drivers of pedestrian presence.**"

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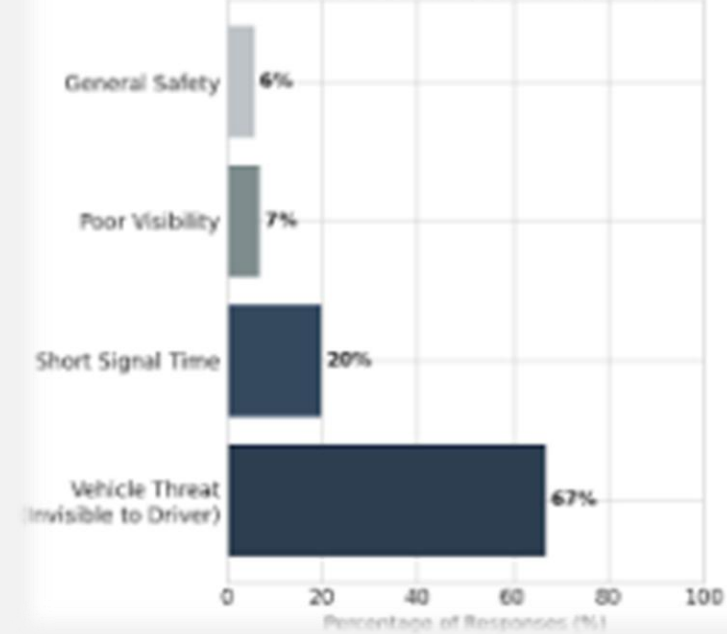
Field Obs. 1: Signal Time Sufficiency  
(Target: Vulnerable Pedestrians)



## "Only 50% Safe"

Half of the observed pedestrians failed to cross within the signal time.

Field Obs. 2: Primary Safety Concerns



## "Fear of Invisibility"

67% of respondents identified "drivers not seeing them" as their top fear.

Public Acceptance: Adaptive Dimming



## "Market Readiness"

Over 80% of citizens agree with the smart dimming & safety system.



# NEED

: A Connected Sensory System

We need an infrastructure that recognizes human presence and environmental hazards instantly.

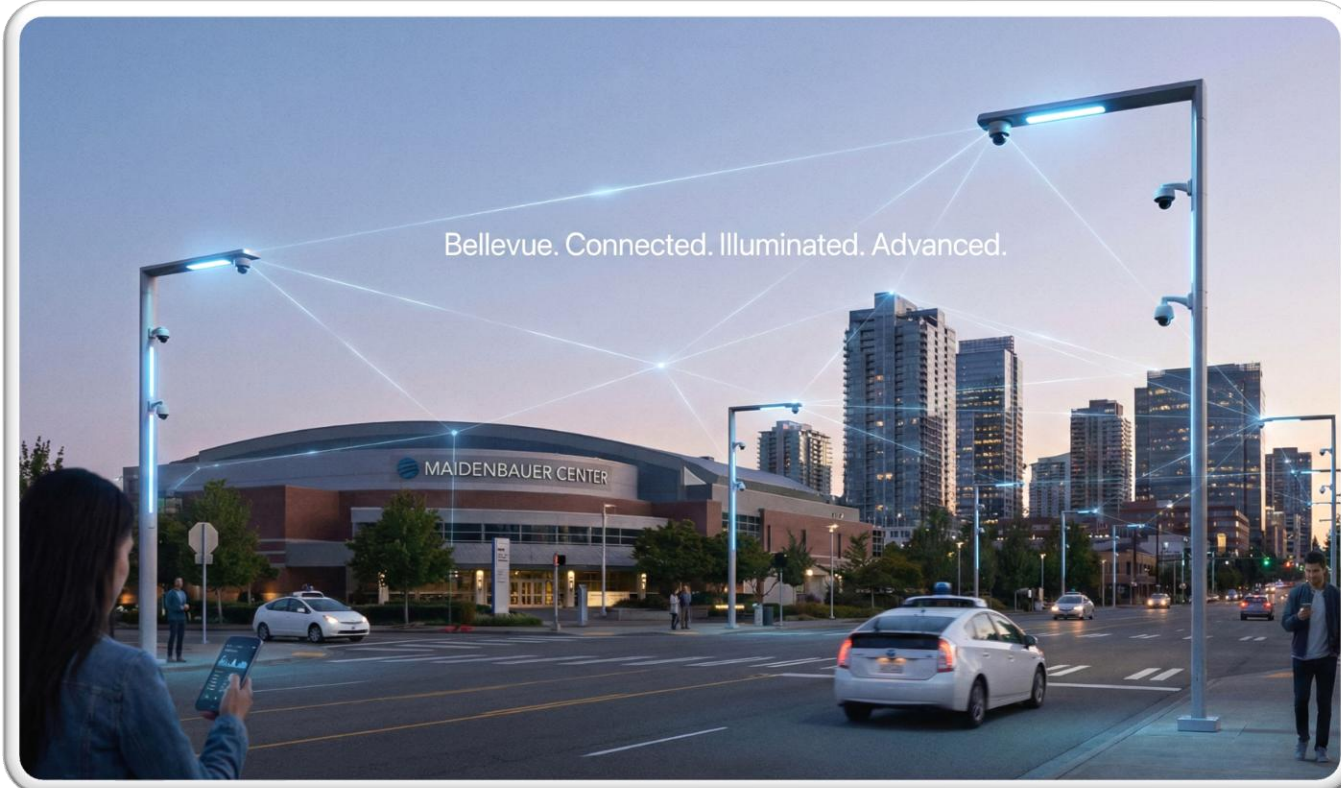
The transition from a simple light source to an active safety system that interacts with citizens.

Don't worry.

**SOLUTION**

**: LightWise Urban SYSTEM**





## 1. "Streetlights are the City's Neural Network"

While streetlights were traditionally seen as just "lighting," they are now evolving into "**Data Collection Hubs (IoT Hubs).**"

- **AS-IS:** Crosswalk Safety Lighting (Safety Only)
- **TO-BE:** LightWise Urban Platform (Urban Data Collection & Control Platform)

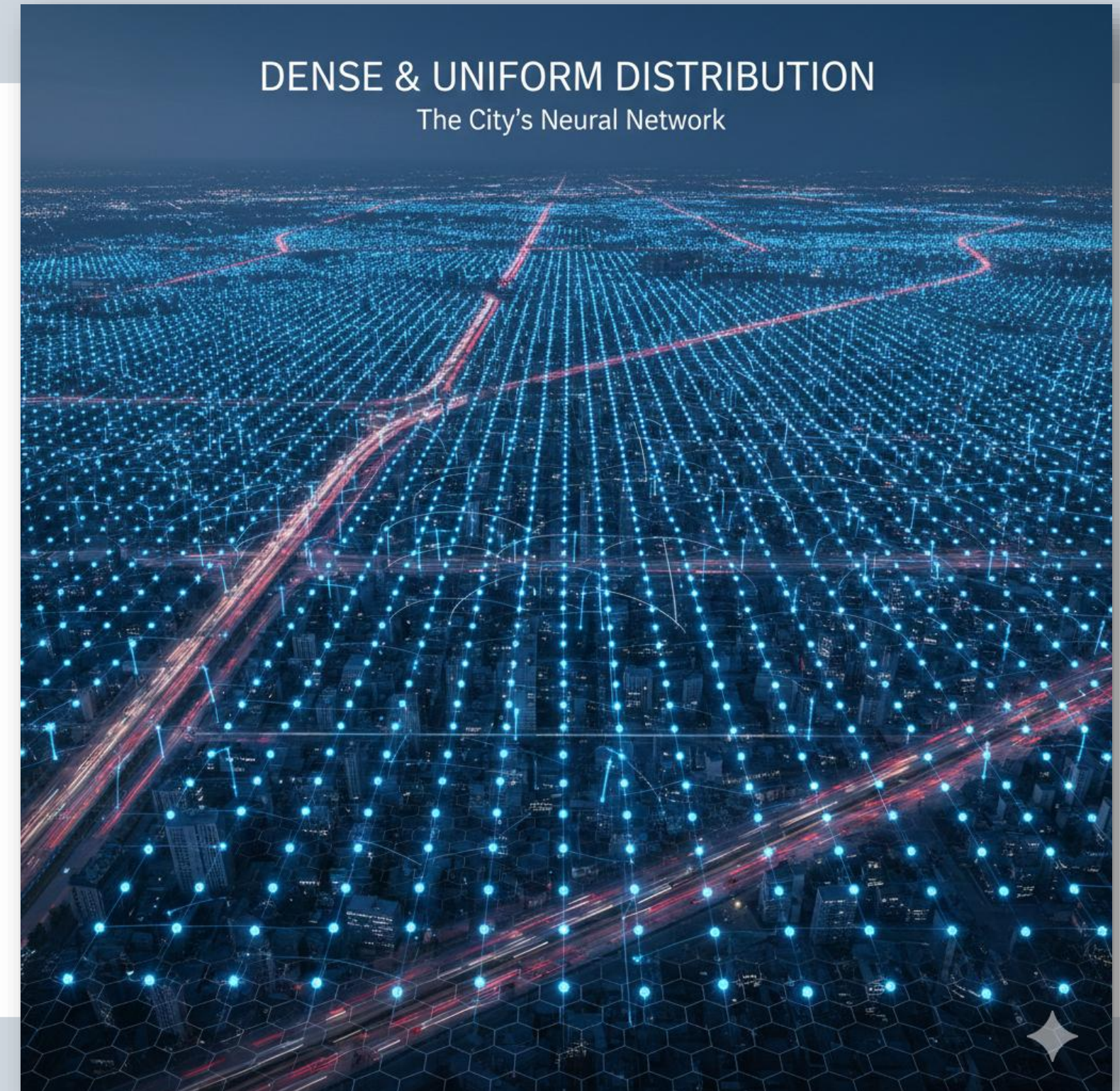
### System Architecture:

- **Input (Sensory Organs):** Infrared/Motion sensors (Pedestrians), Weather sensors (Rain/Snow), Illuminance sensors (Day/Night)
- **Core (Brain):** Analyzes collected data to determine situational context.
- **Output (Action):**
  - Safety:** In-ground crosswalk lighting (Pedestrian Protection)
  - Eco:** Smart Dimming (Energy Savings)
  - Admin:** Auto-fault detection & Dashboard (Maintenance Efficiency)

## WHY Streetlights are the Optimal Infrastructure for a 'City Neural Network'?

### Dense and Uniform Distribution

: strategically placed along sidewalks and roads – the most active interfaces of a city. This provides an unparalleled density for precise data collection. They are everywhere, providing the perfect points for data collection. Streetlights are ubiquitous in urban areas,



## ① Crosswalk Solution (Safety)

**Context:** Addressing the data regarding "72% of accidents occurring at intersections" and "53% of drivers failing to yield."

**Key Appeal:** "The platform's primary function is to solve the most urgent issue: '**Pedestrian Safety.**' When the sensors detect a pedestrian, the system immediately activates the in-ground lighting."





"When no one is around, the streetlight stays in **low-power mode** to maximize energy efficiency."



"As someone approaches, the light **brightens automatically** to ensure pedestrian safety."



"Once the person passes, the light **dims back** to its original level to save power."

② Smart Dimming Service (Sustainability)

**Bellevue's Needs:** The City of Bellevue places high importance on 'Environmental Stewardship' and 'Climate Targets.'

**Key Appeal:** "Keeping streetlights at 100% intensity at 3 AM when no one is around is wasteful. Our sensors reduce brightness to 30% when no pedestrians are present, conserving energy and minimizing **Light Pollution**. This directly aligns with Bellevue's eco-friendly goals."



③ Fault Detection Dashboard (Maintenance Efficiency)

**Bellevue's Needs:** Referencing the "Digital Infrastructure" section (Page 8) of the report, Bellevue aims to manage city assets digitally.

**Key Appeal:** "Previously, the city relied on citizen reports to identify broken lights. LightWise identifies malfunction locations in real-time via our dashboard. This reduces unnecessary field maintenance trips and **drastically cuts administrative costs.**"



System Status

**Optimal**



Today's Energy Savings

**1,245kWh**



Pedestrian Safety Actions

**326**



Faults Detected

**2**



### Safety Insights

Intersection Incident Reduction Goal



Pedestrian Detection & Light Activation (Last 24h)

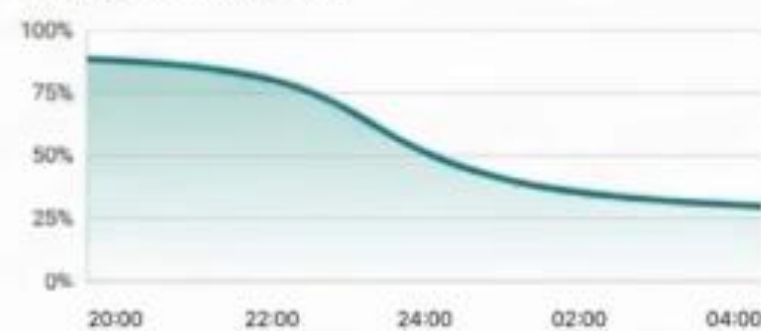


Recent Safety Alerts

- 10:58:57 AM NE 8th St: Activation (Pedestrian)
- 10:53:58 PM NE 8th St: Activation (Pedestrian)
- 10:58:07 PM NE 8th St: Activation (Pedestrian)
- 10:53:50 PM NE 8th St: Activation (Pedestrian)

### Maintenance Efficiency

Energy Efficiency Trend



Smart Dimming Status



Light Pollution Reduction: 85%



### Eco-Sustainability & Dimming

Real-time Fault Detection



Open Maintenance Tickets

- Open Maintenance Status
- Open Maintenance Initiated
- Open Maintenance Fault
- Open Maintenance Status

Cost Savings Estimate:

**\$15k YTD**

Summary and annual cost savings estimate: \$15k YTD

